05

Percentages and applications

Part II · Parts and proportions

CHAPTER 5 · OPENER

Percentages and applications

Percentages express change and comparison relative to a whole; identifying that whole makes financial and fairness decisions more reliable.

ENGAGEMENT

Two shops advertise different discount sequences. Which offer is genuinely cheaper, and what must a price card make clear?

Learning goals

- Convert between fractions, decimals and percentages.
- Find percentages and percentage changes.
- Use multipliers for successive changes.
- Interpret discounts, taxes, costs and profit in a budget.

MYP lens

Key concept: Change

Related concepts: Quantity, Proportion

Global context: Fairness and development

ATL focus. In order for a student to think critically, students must identify the reference quantity, test a claim with calculations and explain the limits of a financial decision.

Language bank: per cent · reference amount · multiplier · percentage change · discount · tax · revenue · cost · gross profit · net profit

CHAPTER 5 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	Find 10% of 90.	READY
R2	Write 1/2 as a decimal and percentage.	READY
R3	Calculate 0.2×50.	READY
R4	Find 60−12.	READY
R5	Write 75/100 in simplest form.	READY

Chapter map

1. **5.1 Per cent and equivalent representations** — Read “per cent” as “out of one hundred”, then connect fractions, decimals and percentages.
2. **5.2 Finding a percentage of a quantity** — Translate “of” into multiplication and identify the reference amount first.
3. **5.3 Percentage increase and decrease** — Find the change from the original amount, or use a single multiplier.
4. **5.4 Discounts, taxes, profit and budgets** — Separate revenue, costs and profit; apply percentages to the correct line of a budget.

> **EXPLAIN HOW YOU KNOW**
>
> For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 5 · MATHEMATICAL DEVELOPMENT

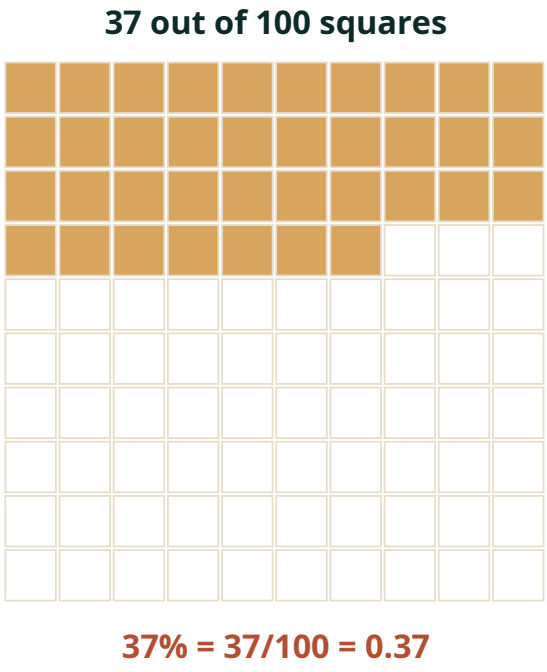
5.1 Per cent and equivalent representations

BIG IDEA

Read “per cent” as “out of one hundred”, then connect fractions, decimals and percentages.

A percentage is a ratio with denominator 100. 37% means 37/100 = 0.37. A percentage can exceed 100 when the amount is more than the reference whole.

To write a fraction as a percentage, find an equivalent fraction out of 100 or multiply by 100%. To write a percentage as a fraction, put it over 100 and simplify.



37 of 100 equal squares: 37/100 = 0.37 = 37%.

WORKED EXAMPLE · CONNECT 3/5, 0.6 AND 60%

Scale the fraction to hundredths.

1. 3/5 = 60/100.
2. 60/100 = 0.60 = 60%.

All three name the same proportion.

CHAPTER 5 · PRACTICE

5.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Percentages compare to a stated whole. 50% of different totals need not be equal amounts.

GUIDED PRACTICE · FIRST STEPS

5.1.1 Write 37%, 5% and 125% as decimals. FLUENCY

5.1.2 Write 3/5, 7/20 and 1/8 as percentages. FLUENCY

5.1.3 Write 24% and 150% as fractions in simplest form. FLUENCY

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.1.4 A student says 0.5% = 0.5. Correct the statement. REASONING

5.1.5 In a group of 40, 18 choose football. What percentage is that? APPLICATION

5.1.6 Can a quantity be 120% of a reference amount? Give an example. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · MATHEMATICAL DEVELOPMENT

5.2 Finding a percentage of a quantity

BIG IDEA

Translate “of” into multiplication and identify the reference amount first.

p% of A means p/100 × A. Useful benchmarks are 10% (divide by 10), 1% (divide by 100), 50% (halve) and 25% (quarter). Build other percentages from these.

For 15% of 240, find 10% = 24 and 5% = 12, then add: 36. Or use the multiplier 0.15 × 240 = 36.

WORKED EXAMPLE · FIND 12.5% OF 320

Use a fraction benchmark.

1. 12.5% = 1/8.
2. 320 ÷ 8 = 40.

40.

CHAPTER 5 · PRACTICE

5.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A percentage is not an amount until its whole is known. 20% of 50 and 20% of 200 are different.

GUIDED PRACTICE · FIRST STEPS

5.2.1 Find 10%, 25% and 15% of 240. FLUENCY

5.2.2 Find 12.5% of 320 and 1% of 750. FLUENCY

5.2.3 Explain two ways to find 35% of 80. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.2.4 A £60 jacket is reduced by 20%. Find the reduction and sale price. APPLICATION

5.2.5 Is 30% of 50 equal to 50% of 30? Explain. REASONING

5.2.6 15% of a number is 27. Find the number. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · MATHEMATICAL DEVELOPMENT

5.3 Percentage increase and decrease

BIG IDEA

Find the change from the original amount, or use a single multiplier.

An increase of p% uses multiplier 1 + p/100; a decrease uses 1 − p/100. The original amount is the reference unless the question says otherwise.

Percentage change = change/original × 100%. A 20% increase followed by a 20% decrease uses 1.2 × 0.8 = 0.96, ending at 96% of the start.



Each percentage uses the current amount as its base.

Each multiplier acts on the current amount: 250 ▒ 200 ▒ 180.

WORKED EXAMPLE · £250 WITH 20% OFF, THEN 10% OFF

Apply each multiplier to the current price.

1. 250 × 0.8 = 200.
2. 200 × 0.9 = 180.
3. A single 30% discount would give 175.

£180; total discount 28%, not 30%.

CHAPTER 5 · PRACTICE

5.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

The second percentage uses a new base. Do not add successive percentage changes unless they apply to the same unchanged whole.

GUIDED PRACTICE · FIRST STEPS

5.3.1 Increase 80 by 15% and decrease 80 by 15%. FLUENCY

5.3.2 A price rises from 50 to 60. Find the percentage increase. FLUENCY

5.3.3 Apply 20% off then 10% off to £250. Compare with 30% off. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.3.4 Increase £100 by 20%, then decrease the result by 20%. Explain the final amount. REASONING

5.3.5 Does reversing 10% off and 20% off change the final price? REASONING

5.3.6 A value falls by 20%. What percentage increase restores the original? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · MATHEMATICAL DEVELOPMENT

5.4 Discounts, taxes, profit and budgets

BIG IDEA

Separate revenue, costs and profit; apply percentages to the correct line of a budget.

A discount reduces a marked price. A tax or tip usually increases a stated base. Revenue is money received; cost is money spent; profit is revenue minus all relevant costs.

For a budget, calculate each line, then add. Whole-item purchases may require rounding up. A lower unit price is not automatically a better decision if extra stock cannot be sold.

WORKED EXAMPLE · A MARKET-STALL BUDGET

Sell 20 honey jars at £8.10, 36 soaps at £3 and 25 jams at £5.

1. Revenue = 162 + 108 + 125 = £395.
2. Goods cost = 20×6 + 36×2.50 + 25×3.20 = £290.
3. Fixed costs = 45 + 18 = £63.

Net profit = 395 − 290 − 63 = £42.

CHAPTER 5 · PRACTICE

5.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Gross profit subtracts goods cost only; net profit also subtracts fixed costs. State which profit you mean.

GUIDED PRACTICE · FIRST STEPS

5.4.1 A £9 item has 10% off. A £4 item has 25% off. Find both sale prices. FLUENCY

5.4.2 A meal costs £40 before 10% tax. Find the total. APPLICATION

5.4.3 Revenue is £395, goods cost £290 and fixed costs £63. Find gross and net profit. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

5.4.4 A shop buys an item for £6 and sells it for £9. Is the profit 50% of cost or of selling price? Calculate both. REASONING

5.4.5 A group needs 23 jars at £6 each. A supplier offers 30 jars at 10% off the £6 cost. Compare total outlay. APPLICATION

5.4.6 Fixed costs of £63 rise by 8%. With revenue £395 and goods cost £290, find the new net profit. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 5 · EXPLORATION

Do two discounts add up?

Offer A: 20% off, then 10% off. Offer B: 30% off. Test both on £250, £40 and £80.

> **INVESTIGATION**
> Criterion B. Use multipliers to test a rule, then explain why it holds for every starting price.

START	A: 20% THEN 10%	B: 30%
£250		
£40		
£80		

EXPLORATION · DO TWO DISCOUNTS ADD UP?

Criterion B. Use multipliers to test a rule, then explain why it holds for every starting price.

B1 Complete the table for £250, £40 and £80. INVESTIGATION

B2 What single discount is equivalent to 20% then 10%? Does order matter? INVESTIGATION

B3 Find the combined discount for 25% then 20%. INVESTIGATION

B4 Is there a distinct unordered pair of whole-number percentage discounts, other than 20% and 10%, giving exactly 28% off? Justify your answer. CHALLENGE

B5 Write a rule for successive discounts a% and b%, and explain the base. INVESTIGATION

CHAPTER 5 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	Find 15% of 200.	CHECK
C2	Increase 50 by 12%.	CHECK
C3	A price falls from 80 to 60. Find the percentage decrease.	CHECK
C4	Find the single multiplier for 10% off then 20% off.	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 5 · APPLICATION

The community market stall

Expected sales: 20 honey jars, 36 soaps and 25 jams. Fixed costs are £45 for the stall and £18 for transport.

REAL-LIFE APPLICATION

Criterion D. Simulated stall data in pounds sterling. Distinguish revenue, goods cost, fixed cost and profit.

PRODUCT	COST EACH	MARKED PRICE	DISCOUNT	EXPECTED SALES
Honey	£6.00	£9.00	10%	20
Soap	£2.50	£4.00	25%	36
Jam	£3.20	£5.00	0%	25

APPLICATION · THE COMMUNITY MARKET STALL

Criterion D. Simulated stall data in pounds sterling. Distinguish revenue, goods cost, fixed cost and profit.

D1 Find each sale price and profit per item. APPLICATION

D2 Find total revenue, goods cost, gross profit and net profit. APPLICATION

D3 Next time, buy 15% more of each item, rounding up to whole items. Find the new goods cost. CHALLENGE

D4 Should the group accept 30 honey jars at 10% off cost instead of the 23 jars needed? Give a financial reason. REASONING

CHAPTER 5 · ASSESSMENT PRACTICE

The fair-price card

MYP ASSESSMENT PRACTICE

Criteria A and C. Design a short price card that shows the original price, discount, final price and one check.

Choose one item priced at £80. Compare “25% off” with “20% off, then 5% off”.

ASSESSMENT PRACTICE · THE FAIR-PRICE CARD

Criteria A and C. Design a short price card that shows the original price, discount, final price and one check.

S1	Calculate both final prices and the better offer.	ASSESSMENT
S2	Write the equivalent single discount for 20% then 5%.	ASSESSMENT
S3	Give a one-sentence warning that prevents a common percentage error.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 5 · MIXED PRACTICE

Mixed practice 5

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 5

Connect ideas. Give units and reasons where requested.

M1	Write 0.08 as a percentage.	FLUENCY
M2	Find 35% of 60.	FLUENCY
M3	Increase 120 by 5%.	FLUENCY
M4	A £50 item has 30% off. Find the sale price.	APPLICATION
M5	A value rises from 40 to 50. Find the percentage increase.	REASONING
M6	Explain why 50% off then 50% off is not free.	REASONING
M7	Revenue £200, costs £145. Find profit and profit as a percentage of revenue.	APPLICATION
M8	After a 10% increase a price is £88. Find the original price.	CHALLENGE
M9	Find the single multiplier for a 10% increase followed by a 10% decrease.	CHALLENGE
M10	A group needs 18 bottles. Packs of 6 cost £4.50. Find the minimum cost.	APPLICATION

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 5 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **9.**
Divide by 10.

R2 **0.5 and 50%.**
Half of 100 is 50.

R3 **10.**
One fifth of 50.

R4 **48.**
Subtract the reduction.

R5 **3/4.**
Divide by 25.

CHECKPOINT

C1 **30.**
10% is 20; 5% is 10.

C2 **56.**
50×1.12.

C3 **25%.**
20/80×100.

C4 **0.72, equivalent to 28% off.**
0.9×0.8.

CHAPTER 5 · ANSWERS AND REASONING

Answers · 5.1 and 5.2

PRACTICE 5.1 · PER CENT

5.1.1 **0.37, 0.05 and 1.25.**
Divide by 100.

5.1.2 **60%, 35% and 12.5%.**
Multiply each fraction by 100%.

5.1.3 **6/25 and 3/2.**
24/100 and 150/100, then simplify.

5.1.4 **0.5% = 0.005. 0.5 = 50%.**
Per cent means divide by 100.

5.1.5 **45%.**
18/40 × 100 = 45.

5.1.6 **Yes. If the reference is 50, then 60 is 120% of it.**
Percentages above 100 represent more than the reference whole.

PRACTICE 5.2 · PERCENTAGE OF

5.2.1 **24, 60 and 36.**
Use a tenth, a quarter, and a tenth plus half a tenth.

5.2.2 **40 and 7.5.**
12.5% = 1/8; 1% means divide by 100.

5.2.3 **28. Use 0.35×80, or 25% (20) + 10% (8).**
Both methods use the same reference whole.

5.2.4 **Reduction £12; sale price £48.**
20% of 60 is 12; subtract from 60.

5.2.5 **Yes, both are 15. Multiplication can be done in either order.**
30/100×50 = 50/100×30.

5.2.6 **180.**
27 ÷ 0.15 = 180.

CHAPTER 5 · ANSWERS AND REASONING

Answers · 5.3 and 5.4

PRACTICE 5.3 · CHANGE

5.3.1 **92 and 68.**
Use multipliers 1.15 and 0.85.

5.3.2 **20%.**
Change 10 divided by original 50, times 100.

5.3.3 **£180 versus £175; the single 30% offer is £5 cheaper.**
250×0.8×0.9 = 180; 250×0.7 = 175.

5.3.4 **£96. The 20% decrease is £24, larger than the original £20 increase.**
The base changed from 100 to 120.

5.3.5 **No. 0.9×0.8 = 0.8×0.9 = 0.72.**
Multiplication is commutative.

5.3.6 **25%.**
From 80 back to 100, the increase is 20/80 = 25%.

PRACTICE 5.4 · MONEY DECISIONS

5.4.1 **£8.10 and £3.00.**
Use multipliers 0.9 and 0.75.

5.4.2 **£44.**
Tax £4; total 40×1.1.

5.4.3 **Gross £105; net £42.**
Gross = 395−290; net = 105−63.

5.4.4 **£3 profit: 50% of cost, 33 1/3% of selling price.**
The denominator changes the percentage.

5.4.5 **23 jars cost £138; 30 at £5.40 cost £162, £24 more.**
The lower unit price requires buying 7 extra jars.

5.4.6 **£36.96.**
New fixed costs 63×1.08 = £68.04; net = 395−290−68.04.

CHAPTER 5 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · DO TWO DISCOUNTS ADD UP?

B1 **A: £180, £28.80, £57.60. B: £175, £28, £56.**
Use 0.8×0.9 = 0.72 and 0.7.

B2 **28%; order does not matter because 0.8×0.9 = 0.9×0.8.**
The product of multipliers is unchanged by order.

B3 **40%.**
0.75×0.8 = 0.60 remains.

B4 **Yes: 4% then 25% (or 25% then 4%), since 0.96×0.75 = 0.72. The pair 10% and 20% also works but is excluded by the question.**
Solve (100−a)(100−b)=7200; factor pairs in range include 96×75 and 80×90.

B5 **Remaining multiplier (1−a/100)(1−b/100); combined discount is 1 minus this product. The second discount uses the reduced price.**
Multiplying multipliers tracks the changing base.

CHAPTER 5 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE COMMUNITY MARKET STALL

D1 **Honey £8.10, profit £2.10; soap £3.00, profit £0.50; jam £5.00, profit £1.80.**
Apply each discount, then subtract cost.

D2 **Revenue £395; goods £290; gross £105; net £42.**
Revenue 162+108+125; goods 120+90+80; fixed 63.

D3 **23 honey, 42 soap, 29 jam; cost £335.80.**
Ceil(20×1.15)=23, ceil(36×1.15)=42, ceil(25×1.15)=29.

D4 **Not on cost alone: £162 versus £138, so £24 more is tied up in stock. Accept only if extra demand or storage justifies it.**
Compare total outlay and risk, not just unit price.

ASSESSMENT PRACTICE · THE FAIR-PRICE CARD

S1 **£60 versus £60.80; 25% off is 80p cheaper.**
80×0.75 = 60; 80×0.8×0.95 = 60.8.

S2 **24%.**
1−0.8×0.95 = 0.24.

S3 **Sample: “The second discount applies to the reduced price, so 20% and 5% do not combine to 25%.”**
The warning identifies the changing base.

CHAPTER 5 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 5

M1 **8%.**
Multiply by 100.

M2 **21.**
0.35×60.

M3 **126.**
120×1.05.

M4 **£35.**
50×0.7.

M5 **25%.**
10/40×100.

M6 **The final price is 25% of the original: 0.5×0.5 = 0.25.**
The second half is taken from the reduced price.

M7 **£55; 27.5%.**
55/200×100.

M8 **£80.**
88÷1.1.

M9 **0.99, a 1% decrease.**
1.1×0.9.

M10 **£13.50 for 3 packs.**
18÷6 = 3; 3×4.50.